Boss Challenge - Paper 08 (Class 7)

Acids, Bases & Salts | Electric Current & its Effects | Algebraic Expressions | Exponents & Powers
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A liquid turns blue litmus red but leaves red litmus unchanged. What can you safely conclude?
   (A) It is basic.
   (B) It is a salt and so always basic.
   (C) It is acidic.
   (D) It is neutral.

2. In the term -7xy, the numerical coefficient is:
   (A) xy
   (B) -7
   (C) 7
   (D) -7xy

3. In a circuit diagram, a long thin line next to a short thick line represents:
   (A) a resistor
   (B) a switch
   (C) a bulb
   (D) a cell

4. The expression 2^5 means:
   (A) 5 x 5
   (B) 2 + 2 + 2 + 2 + 2
   (C) 2 multiplied by itself five times (2 x 2 x 2 x 2 x 2)
   (D) 2 x 5

5. Phenolphthalein stays colourless in liquid X. When X is added to a pink phenolphthalein-base mixture, the pink fades. X is most likely:
   (A) an acid
   (B) pure water
   (C) a strong base
   (D) a basic salt

6. Which pair are LIKE terms?
   (A) 5x^2 and 5x
   (B) 2xy and 2x
   (C) 3a and 3b
   (D) 3ab and -7ba

7. Two or more cells joined together form a:
   (A) battery
   (B) filament
   (C) fuse
   (D) switch

8. The value of 3^4 is:
   (A) 27
   (B) 12
   (C) 64
   (D) 81

9. An ant's sting injects formic acid into the skin. The best quick remedy from the kitchen is to rub on:
   (A) lemon juice
   (B) vinegar
   (C) common salt solution
   (D) baking soda (a mild base)

10. If $x = 2$ and $y = 3$, the value of $2x^2 - y$ is:
   (A) -1
   (B) 5
   (C) 10
   (D) 13

11. To make a working battery, cells are joined so that:
   (A) the positive terminal of one cell connects to the negative terminal of the next
   (B) the negative terminal joins the negative terminal
   (C) the terminals are left unconnected
   (D) the positive terminal joins the positive terminal

12. Using the laws of exponents, $3^2 \times 3^3$ equals:
   (A) 3^5
   (B) 3^1
   (C) 9^5
   (D) 3^6

13. Which group contains only acids?
   (A) curd, soap solution, orange
   (B) lemon, vinegar, milk of magnesia
   (C) ant sting, lime water, spinach
   (D) lemon (citric), vinegar (acetic), tamarind (tartaric)

14. Sohan is 5 years older than twice Ravi's age. If Ravi is r years old, Sohan's age is:
   (A) $5r + 2$
   (B) $r + 5$
   (C) $2r + 5$
   (D) $2(r + 5)$

15. When current passes through a wire and makes it hot, this is called the:
   (A) heating effect of current
   (B) chemical effect of current
   (C) magnetic effect of current
   (D) lighting effect of current

16. Using the laws of exponents, $5^7 / 5^4$ equals:
   (A) $5^{(7/4)}$
   (B) 1^3
   (C) 5^3
   (D) 5^{11}

17. A drop of soap solution on a yellow turmeric stain turns the stain red. This shows soap solution is:
   (A) neutral
   (B) basic
   (C) acidic
   (D) a salt with no effect

18. How many terms are in the expression $4x^2 - 3xy + 7$?
(A) 1
(B) 3
(C) 4
(D) 2

19. Equal currents pass through a thick copper wire and a thin nichrome wire. Which gets hotter?
(A) the thick copper wire
(B) both heat equally
(C) neither heats at all
(D) the thin nichrome wire

20. The value of $(2^3)^2$ is:
(A) 2
(B) 64
(C) 12
(D) 32

21. When an acid reacts completely with a base, the products are always:
(A) hydrogen gas and a salt
(B) a salt and water
(C) two salts
(D) a base and water

22. Subtract $(3a - 2b)$ from $(5a + b)$.
(A) $-2a - 3b$
(B) $8a - b$
(C) $2a - 3b$
(D) $2a + 3b$

23. The filament of an electric bulb glows mainly because:
(A) the glass cover gets hot
(B) the bulb is filled with water
(C) current heats the thin high-resistance filament until it gives out light
(D) the switch becomes a magnet

24. Any non-zero number raised to the power 0 equals:
(A) 1
(B) undefined
(C) 0
(D) the number itself

25. A farmer's field is too acidic for crops. The correct substance to add to the soil is:
(A) vinegar
(B) common salt
(C) slaked lime (a base)
(D) an acid fertiliser

26. Simplify: $7x - 3x + 2 - 5$.
(A) $4x - 3$
(B) $10x - 3$
(C) $4x + 3$
(D) $4x - 7$

27. A fuse protects an electrical circuit because it:
    (A) increases the current in the circuit
    (B) stores electric charge for later
    (C) melts and breaks the circuit when the current becomes too large
    (D) makes the bulb glow brighter

28. The value of $(-1)^{100}$ is:
    (A) -1
    (B) 1
    (C) -100
    (D) 100

29. China rose indicator turns a solution green. The solution must be:
    (A) acidic
    (B) always a salt
    (C) neutral
    (D) basic

30. The expression $3(2x + 4)$ is equal to:
    (A) $6x + 12$
    (B) $6x + 4$
    (C) $6x + 7$
    (D) $5x + 7$

31. A fuse wire is deliberately made of a metal that melts easily so that:
    (A) the current in the circuit increases
    (B) the wire becomes a permanent magnet
    (C) it breaks the circuit before the appliance's own wires overheat
    (D) the bulb glows brighter

32. The value of $(-1)^{101}$ is:
    (A) 101
    (B) -101
    (C) -1
    (D) 1

33. Why must a factory neutralise its acidic waste before letting it flow into a river?
    (A) to remove the acid's harmful effect and protect aquatic life
    (B) because salts always evaporate
    (C) to make the water taste sweet
    (D) to increase the acid's strength

34. In $5x^2 - 2x + 9$, the constant term is:
    (A) x
    (B) -2
    (C) 5
    (D) 9

35. A compass needle near a wire deflects only when the switch is ON. This shows that:
    (A) the wire is charged like a rubbed balloon
    (B) the compass is faulty
    (C) an electric current produces a magnetic effect
    (D) current produces only heat

36. The number 47000 written in standard form is:
   (A) 4.7 x 10^4
   (B) 4.7 x 10^3
   (C) 0.47 x 10^5
   (D) 47 x 10^3

37. Fresh milk is almost neutral, but on standing it slowly turns sour. The best explanation is that:
   (A) an acid (lactic acid) forms in it
   (B) a base forms in it
   (C) it turns into pure water
   (D) a salt forms and makes it sour

38. The perimeter of a rectangle with length l and breadth b is given by the expression:
   (A) l x b
   (B) 2(l + b)
   (C) 2l + b
   (D) l + b

39. A coil of wire wound around an iron nail acts as a magnet ONLY while current flows. This device is a/an:
   (A) permanent magnet
   (B) electromagnet
   (C) fuse
   (D) cell

40. The number 5.2 x 10^3 equals:
   (A) 5.2000
   (B) 520
   (C) 5200
   (D) 52000

41. An antacid tablet relieves acidity in the stomach because it:
   (A) is a mild base that neutralises excess stomach acid
   (B) is a neutral salt that absorbs water
   (C) adds more acid to digest food
   (D) is a strong acid that burns the germs

42. Joined squares in a row need 3n + 1 matchsticks for n squares. How many sticks are needed for 10 squares?
   (A) 31
   (B) 30
   (C) 34
   (D) 40

43. Which change makes an electromagnet STRONGER?
   (A) switching off the current
   (B) removing the iron core
   (C) using fewer turns
   (D) increasing the number of turns in the coil

44. Which is greater, 2^5 or 5^2?
   (A) 5^2
   (B) they are equal
   (C) 2^5
   (D) they cannot be compared

45. Which liquid is most likely to leave BOTH red and blue litmus unchanged?
   (A) distilled water
   (B) lemon juice
   (C) soap solution
   (D) baking soda solution

46. Which of the following is a MONOMIAL?
   (A) $a + b + c$
   (B) $x + y$
   (C) $2x - 3$
   (D) $-6x^2y$

47. The main advantage of an electromagnet over a permanent magnet is that:
   (A) it cannot attract iron
   (B) it never needs any current
   (C) its magnetism can be switched on and off
   (D) it is always weaker than any magnet

48. $(2 \times 3)^2$ is equal to:
   (A) $2 \times 3^2 = 18$
   (B) $2^2 \times 3^2 = 36$
   (C) $2^2 \times 3 = 12$
   (D) $2 \times 3 \times 2 = 12$

49. You have one bottle of dilute acid and one of dilute base, but ONLY red litmus paper. Which can you definitely identify?
   (A) both, immediately
   (B) the base, because it turns red litmus blue
   (C) the acid, because it turns red litmus blue
   (D) neither, you always need blue litmus

50. The factors of the term $5xy$ are:
   (A) $5x$ and $5y$
   (B) 5, x and y
   (C) 5 and xy only
   (D) x and y only

51. An electric bell works mainly using the ____ effect of current.
   (A) magnetic
   (B) chemical
   (C) cooling
   (D) heating

52. Expressed as a product of prime powers, 72 equals:
   (A) $2^4 \times 3$
   (B) $2^3 \times 3^3$
   (C) $2^2 \times 3^3$
   (D) $2^3 \times 3^2$

53. Equal strengths of an acid and a base are mixed to give a neutral solution. A little MORE acid is then added. The mixture becomes:
   (A) turned into pure salt only
   (B) acidic
   (C) basic
   (D) still perfectly neutral

54. If $a = -2$, the value of $a^2 + 3a$ is:
   (A) 10
   (B) 2
   (C) -2
   (D) -10

55. If the switch in a single-loop circuit is left OPEN, the bulb will:
   (A) not glow, because the circuit is incomplete
   (B) glow brighter than before
   (C) glow dimly
   (D) become a magnet

56. Using the laws of exponents, $10^3 \times 10^2 \times 10$ equals:
   (A) 10^7
   (B) 100^6
   (C) 10^6
   (D) 10^5

57. Many toothpastes are mildly basic. The main reason is to:
   (A) dissolve the enamel faster
   (B) make the mouth strongly acidic
   (C) add sugar to feed bacteria
   (D) neutralise the acid made by mouth bacteria

58. Add the expressions $2x^2 + 3x$ and $x^2 - x + 4$.
   (A) $3x^2 + 4x + 4$
   (B) $3x^2 + 2x + 4$
   (C) $3x^2 + 2x - 4$
   (D) $2x^2 + 2x + 4$

59. In a circuit two bulbs are connected one after another (in series). If one bulb fuses, the other bulb will:
   (A) glow much brighter
   (B) turn into a fuse
   (C) also stop glowing, because the circuit is broken
   (D) be completely unaffected

60. The value of $7^5 / 7^5$ is:
   (A) 7
   (B) 0
   (C) 7^{10}
   (D) 1

61. Which of these is a base?
   (A) caustic soda (sodium hydroxide)
   (B) common salt
   (C) citric acid
   (D) vinegar

62. In the expression $x^2 - x$, the coefficient of x is:
   (A) -1
   (B) x
   (C) 1
   (D) 0

63. Adding one more cell (connected correctly) to a torch usually makes the bulb:
   (A) exactly the same
   (B) magnetic instead of bright
   (C) dimmer
   (D) brighter, because more current flows

64. A distance is written as 9 x 10^7 km. Written in full, this is:
   (A) 9,000,000 km
   (B) 79,000,000 km
   (C) 90,000,000 km
   (D) 900,000,000 km

65. Common salt is dissolved in water. The solution will be:
   (A) able to turn blue litmus red
   (B) strongly acidic
   (C) neutral
   (D) strongly basic

66. A hall has c chairs (4 legs each) and t stools (3 legs each). The total number of legs is:
   (A) $3c + 4t$
   (B) $4c + 3$
   (C) $7(c + t)$
   (D) $4c + 3t$

67. The heating element of an electric room heater is made of nichrome because nichrome:
   (A) is naturally magnetic
   (B) has a high resistance and a high melting point
   (C) is a poor conductor that blocks current completely
   (D) melts very easily

68. Which of the following equals 64?
   (A) 2^5
   (B) 2^6
   (C) 4^4
   (D) 3^4

69. Litmus, the most common natural indicator, is obtained from:
   (A) lichens
   (B) rose petals
   (C) only red cabbage
   (D) turmeric roots

70. Which two terms CANNOT be added together (they are unlike)?
   (A) x and $4x$
   (B) $-3ab$ and ab
   (C) $6x^2$ and $6x$
   (D) $2y$ and $5y$

71. Why should you NEVER touch an electric switch with wet hands?
   (A) water cools the switch and breaks it
   (B) water lets electricity pass through you easily, risking a dangerous shock
   (C) wet hands repel electric current
   (D) wet hands turn into magnets

72. Since $a^3 = a \times a \times a$, the product $a^3 \times a^2$ is the same as:
   (A) a^6
   (B) a^9
   (C) a^1
   (D) $a \times a \times a \times a \times a = a^5$

73. Plants are dying in a strongly BASIC soil. Which addition best moves the soil towards neutral?
   (A) organic manure / compost
   (B) slaked lime
   (C) baking soda
   (D) caustic soda

74. The cost in rupees of x pencils at Rs 4 each plus one Rs 10 eraser is $C = 4x + 10$. The cost of 7 pencils and one eraser is:
   (A) Rs 38
   (B) Rs 40
   (C) Rs 28
   (D) Rs 48

75. An electromagnet uses a soft iron core rather than a steel core because soft iron:
   (A) loses its magnetism quickly when the current is switched off
   (B) conducts no electric current
   (C) cannot be magnetised at all
   (D) stays a magnet forever after one use

76. Six lakh forty thousand (6,40,000) written in standard form is:
   (A) 64×10^4
   (B) 0.64×10^6
   (C) 6.4×10^4
   (D) 6.4×10^5

77. Phenolphthalein is pink in a solution. Which single action will most likely turn it colourless?
   (A) cooling the solution
   (B) adding common salt
   (C) adding more base
   (D) adding enough acid

78. Simplify: $3a + 4b - a + 2b$.
   (A) $4a + 6b$
   (B) $2a + 6b$
   (C) $2a - 6b$
   (D) $2a + 2b$

79. Which device works mainly on the MAGNETIC effect of current?
   (A) an electric bulb
   (B) an electromagnet crane lifting scrap iron
   (C) an electric iron (press)
   (D) a room heater

80. $(5^2)^3$ has the same value as:
   (A) 5^5
   (B) 5^8
   (C) 5^6
   (D) 5^{23}

81. Pouring vinegar on baking soda makes bubbles of gas. This tells us that vinegar is:
   (A) neutral
   (B) basic
   (C) acidic
   (D) a metal

82. Which expression is a BINOMIAL?
   (A) 2x - 5
   (B) 3xy
   (C) a + b + c
   (D) 7

83. Which appliance works mainly on the HEATING effect of current?
   (A) an electromagnet crane
   (B) a compass near a wire
   (C) an electric bell
   (D) an electric toaster

84. Without multiplying it all out, the sign of (-2)^6 is:
   (A) cannot be decided without full multiplication
   (B) negative
   (C) positive
   (D) zero

85. Which one of these statements about acids and bases is correct?
   (A) Bases turn blue litmus red.
   (B) Acids turn blue litmus red; bases turn red litmus blue.
   (C) Neutral liquids turn both litmus papers red.
   (D) Acids turn red litmus blue.

86. If p = 3, the value of 2p^2 is:
   (A) 18
   (B) 9
   (C) 12
   (D) 36

87. In a circuit diagram, an OPEN switch means the switch is:
   (A) a cell that supplies current
   (B) OFF, so current cannot flow
   (C) ON, so current flows freely
   (D) a fuse that has melted

88. Which is greater, 3^4 or 4^3?
   (A) they are equal
   (B) 4^3
   (C) they cannot be compared
   (D) 3^4

89. Lime water turns red litmus blue. This tells us that lime water is:
   (A) an acid
   (B) neutral
   (C) a strong acid like a lemon
   (D) a base

90. A taxi charges Rs 30 fixed plus Rs 12 per km. For a ride of d km the fare in rupees is:
- (A) 30d + 12
- (B) 42d
- (C) 12d + 30
- (D) 12 + 30d

91. A torch bulb will not light even with a brand-new bulb and the switch closed. The MOST likely cause is:
- (A) the bulb is too new to work
- (B) the connecting wire is too short
- (C) there is too much magnetism in the wire
- (D) the cells are used up, so no current flows

92. Simplify 2^8 / (2^3 x 2^2).
- (A) 2^(8/5)
- (B) 2^3 = 6
- (C) 2^3 (= 8)
- (D) 2^13

93. Solution X turns china rose magenta; solution Y turns turmeric red. X and Y are respectively:
- (A) both acids
- (B) both bases
- (C) a base and an acid
- (D) an acid and a base

94. On collecting like terms, 5xy + 3x - 2xy + x equals:
- (A) 3xy + 3x
- (B) 7xy + 4x
- (C) 3xy + 4x
- (D) 3xy + 2x

95. An appliance normally draws a working current of about 5 A. For safety, the fuse fitted to it should melt at a current that is:
- (A) a little above the normal working current (just over 5 A)
- (B) far below the normal working current
- (C) exactly zero amperes
- (D) as high as possible so it never melts

96. A number in standard form is written as k x 10^n. The value of k must satisfy:
- (A) k is always greater than 10
- (B) 1 <= k < 10
- (C) k is exactly 10
- (D) k can be any whole number

97. A factory releases BASIC waste water. To make it safe before release, the cheapest correct choice is to add a/an:
- (A) common salt
- (B) stronger base
- (C) only fresh water
- (D) acid

98. For the cost expression 8n (n notebooks at Rs 8 each), which statement is correct?
- (A) Both 8 and n are constants.
- (B) 8 is the fixed price per notebook (a constant) and n is the variable number of notebooks.
- (C) 8 is the number of notebooks and n is the price.
- (D) Both 8 and n are variables.

99. In a simple circuit, conventional current is taken to flow through the wire starting from the:
   (A) negative terminal of the cell
   (B) positive terminal of the cell
   (C) both terminals at once
   (D) middle of the connecting wire

100. Light travels about 3 x 10^5 km in one second. In 2 seconds it travels:
   (A) 3 x 10^25 km
   (B) 6 x 10^5 km
   (C) 3 x 10^10 km
   (D) 6 x 10^10 km